

1.11 Project Management Plan

The project will be divided into the following phases:

Phase	Main Activities	Output
Planning	Define objectives, scope and schedule	Project plan
Requirements Analysis	Gather and document requirements	SRS document
Design	Create diagrams, database and interfaces	Design document
Implementation	Develop system modules	Working system
Testing	Test features and error cases	Test report
Deployment	Install the application	Deployed system
Maintenance	Fix errors and add improvements	Updated versions

1.13 Designing Phase

The designing phase will convert requirements into a technical plan.

1.13.1 Interface Design

The main interfaces will include:

- Login screen
- Student dashboard
- Hostel-application form
- Application-status page
- Admin dashboard
- Room-management screen
- Bed-allotment screen
- Complaint screen
- Fee-status screen
- Visitor screen
- Reports screen

1.13.2 Database Design

The database will contain tables for students, hostels, rooms, beds, applications, allotments, complaints, payments, visitors and notices.

1.13.3 UML Diagrams

The following diagrams will be prepared:

- Use-case diagram
- Activity diagrams
- Sequence diagrams
- Class diagram

- Component diagram
- Entity-Relationship Diagram

Detailed UML diagrams will be provided in Chapters 2 and 3.

1.14 Implementation Phase

During implementation, developers will create the database and system interfaces.

The implementation steps are:

1. Create the database.
 2. Develop login and role-based access.
 3. Develop student profiles.
 4. Develop hostel applications.
 5. Develop hostel, room and bed records.
 6. Develop room-allotment functions.
 7. Develop complaint handling.
 8. Develop fee management.
 9. Develop visitor records.
 10. Develop reports.
 11. Connect all modules.
 12. Test the system.
-

4.2 Test Plan

The test plan defines the testing activities, scope and expected results.

The Hostel Management System will be tested in the following stages:

4.2.1 Unit Testing

Unit testing checks individual functions or modules separately.

Examples:

- Login validation
- Application form validation
- Bed-availability check
- Complaint submission
- Payment-status update

4.2.2 Integration Testing

Integration testing checks whether connected modules work correctly together.

Examples:

- Approved application with bed allotment
- Bed allotment with occupancy update
- Complaint submission with staff dashboard
- Payment verification with student fee status
- Visitor entry with resident record

4.2.3 System Testing

System testing checks the complete application as a whole.

It verifies that users can perform their tasks from login to completion.

4.2.4 Security Testing

Security testing checks whether unauthorized users are blocked from protected functions.

Examples:

- Student access to admin dashboard
- Security staff access to payment records
- Invalid login attempts
- Session timeout

4.2.5 User-Acceptance Testing

User-acceptance testing checks whether the application is suitable for students and hostel staff.

Users will test:

- Navigation
- Form clarity
- Status messages
- Reports
- Dashboard information
- Mobile responsiveness

4.7 Security Testing

Security testing will verify the following controls:

Security Area	Test
Authentication	Invalid username and password are rejected
Authorization	Users cannot access functions outside their role
Password Storage	Passwords are not stored as plain text
Session Management	Inactive sessions expire
Input Validation	Invalid data and unsafe input are rejected
File Upload	Unsupported files are blocked
Audit Logging	Important administrative actions are recorded
Data Privacy	Students cannot view other students' private records
Backup	Database backup can be created and restored

4.8 Performance Testing

Performance testing will evaluate normal system response.

Test	Expected Result
Open login page	Page loads without unnecessary delay
Search student record	Correct record appears quickly
Filter applications	Filtered list appears correctly
Generate occupancy report	Report is displayed within a reasonable time
Submit hostel application	Confirmation appears after successful submission
Open dashboard	Summary values are displayed correctly

The exact response-time target may be defined after deployment conditions are known.

4.9 User-Acceptance Testing

User-acceptance testing will be performed with sample users.

4.9.1 Student Evaluation

Students will check whether they can:

- Log in easily
- Submit an application
- Track status
- Submit complaints
- Read notices
- Check fee status

4.9.2 Staff Evaluation

Hostel staff will check whether they can:

- Search applications
- Review student records
- Update status
- Allocate beds
- Verify payments
- Generate reports

4.9.3 Security-Staff Evaluation

Security staff will check whether they can:

- Record visitor entries
- Record visitor exits
- Search visitor records

4.9.4 Acceptance Criteria

The system will be accepted if:

1. Core functions work correctly.
2. Duplicate bed allotments are prevented.
3. Unauthorized access is blocked.
4. Required reports are generated accurately.
5. Users can navigate the system without major difficulty.
6. No critical errors remain unresolved.

4.10 Sample Test Summary Report

Category	Total Test Cases	Passed	Failed	Remarks
Authentication	2	2	0	Login works correctly
Hostel Application	3	3	0	Validation works correctly
Allotment	3	3	0	Duplicate allotment prevented
Renewal and Transfer	2	2	0	Requests are stored correctly
Complaint Management	3	3	0	Status updates work correctly
Fee Management	2	2	0	Payment validation works
Visitor Management	3	3	0	Entry and exit records work
Notices and Reports	3	3	0	Reports and notices display correctly
Security	4	4	0	Unauthorized access is blocked
Total	25	25	0	System ready for final review

The values in this table are sample results. Actual values should be updated after real testing.
